

Arin Balyan

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Experience

SCG Foundation | AI/Full-Stack Engineer

Nov 2025 – Apr 2026

FastAPI, Python, Vue.js, LangChain, LangGraph, PostgreSQL, Docker, AWS, OpenAI API

Remote — Full-time

- Led end-to-end development of a production AI platform, collaborating directly with stakeholders on system architecture, feature design, deployment strategy, and release planning.
- Built and maintained LangChain/LangGraph-based AI agent pipelines with retrieval workflows, tool orchestration, response validation, and multi-step reasoning integrations.
- Integrated and optimized LLM workflows using OpenAI APIs and Anthropic SDKs, improving response quality, reliability, and runtime efficiency across production use cases.
- Handled production debugging, infrastructure scaling, and deployment optimization, supporting stable performance during rapid traffic growth and continuous feature rollouts.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL

AI / ML: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Hugging Face

LLM / Agentic AI: LangChain, LangGraph, RAG, Prompt Engineering

DevOps / Cloud: Docker, AWS, CI/CD, Git, Cloudflare

Backend: FastAPI, Node.js, REST APIs

Databases: PostgreSQL, Redis

Tools: Linux, Kaggle, Streamlit

Projects

JobHunter | AI-Powered Job Automation Pipeline

Feb 2026 – Present

FastAPI, Python, LangChain, GitHub Actions, Cron, SMTP, OpenAI API

[GitHub](#)

- Built a fully automated job scraping pipeline targeting **LinkedIn, Indeed, and Naukri**, aggregating **10,000+ listings daily** across configurable role-based queries — orchestrated via cron and deployed on **GitHub Actions CI/CD** running twice daily without manual intervention.
- Implemented an LLM-based job scoring engine that evaluates each listing against the target role, ranking and filtering high-confidence matches from raw scrape output — reducing noise by **~90%** and surfacing only actionable opportunities.
- Designed a context-aware AI email system that drafts and sends personalized outreach emails for top-scored listings, injecting live job metadata (title, company, description) alongside user-defined profile context from a `context.md` file — resulting in **fully tailored, role-specific emails** with zero manual writing.
- Exposed pipeline controls and job logs via a **FastAPI** backend, enabling real-time status checks, score thresholds, and email trigger configuration through a clean REST interface.

Image Enhancement Web Service | AI Photo Processing Pipeline

Jan 2026 – Mar 2026

Python, FastAPI, PyTorch, Real-ESRGAN, GFPGAN, OpenCV

[GitHub](#)

- Engineered a production-ready service combining **Real-ESRGAN** (4× super-resolution), **GFPGAN** face restoration, and adaptive white-balance/exposure correction.
- Processed high-resolution images with an optimized inference pipeline, reducing average processing time and enabling efficient batch workloads.
- Developed an automated image analysis pipeline to intelligently detect and apply required enhancements; supports **batch processing** and **Google Drive** integration via FastAPI.

EmotionML | Real-Time Emotion Recognition Platform

Jul 2025 – Aug 2025

PyTorch, MobileNetV2, FastAPI, React.js, OpenCV

[GitHub](#)

- Trained **9 deep learning models** on **35,000+ images** (FER-2013 & AffectNet); achieved **98.99% on CK+48** (controlled dataset) and **72.4% on FER-2013** (real-world dataset), outperforming baselines by **12%**.
- Optimised MobileNetV2 reducing model size **60%** (23MB → 9MB) and inference time **40%** (250ms → 150ms), enabling edge deployment on 512MB RAM devices.
- Developed FastAPI backend for real-time inference, enabling efficient image processing workflows with optimized model integration.

Education

Bachelor of Technology — Computer Science (AI & ML)

Aug 2022 – May 2026

Vellore Institute of Technology

CGPA: 8.45 / 10

Courses & Certifications

Applied Machine Learning in Python — University of Michigan

Score: 94%

MERN Full-Stack Developer Certification — Ethnus

Score: 90%

Achievements & Leadership

- Top 25 Global** among **500+ teams** — Zelestra × AWS Hackathon; Designed and deployed an ML solution end-to-end in **48 hours**.
- GSSoC Open-Source Contributor:** 30+ merged PRs across 20+ repositories including ML projects and GenAI projects.